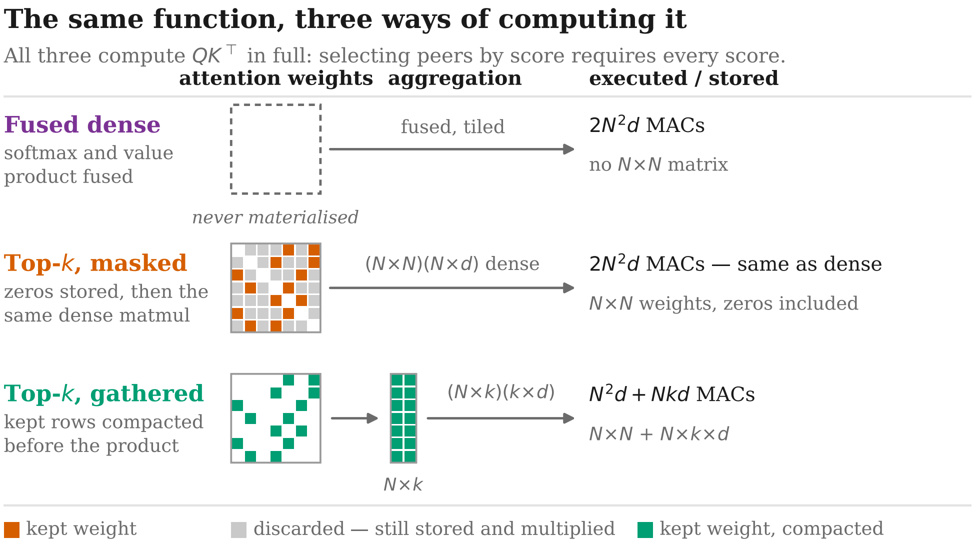

The same function, three ways of computing it
All three compute $QK^\top$ in full: selecting peers by score requires every score.
attention weights
aggregation
executed / stored
Fused dense
softmax and value product fused
never materialised
fused, tiled
$2N^2d$ MACs
no $N\times N$ matrix
Top-k, masked
zeros stored, then the same dense matmul
$(N\times N)(N\times d)$ dense
$2N^2d$ MACs — same as dense
$N\times N$ weights, zeros included
Top-k, gathered
kept rows compacted before the product
$N\times k$
$(N\times k)(k\times d)$
$N^2d + Nkd$ MACs
$N\times N$ + $N\times k\times d$
kept weight
discarded — still stored and multiplied
kept weight, compacted